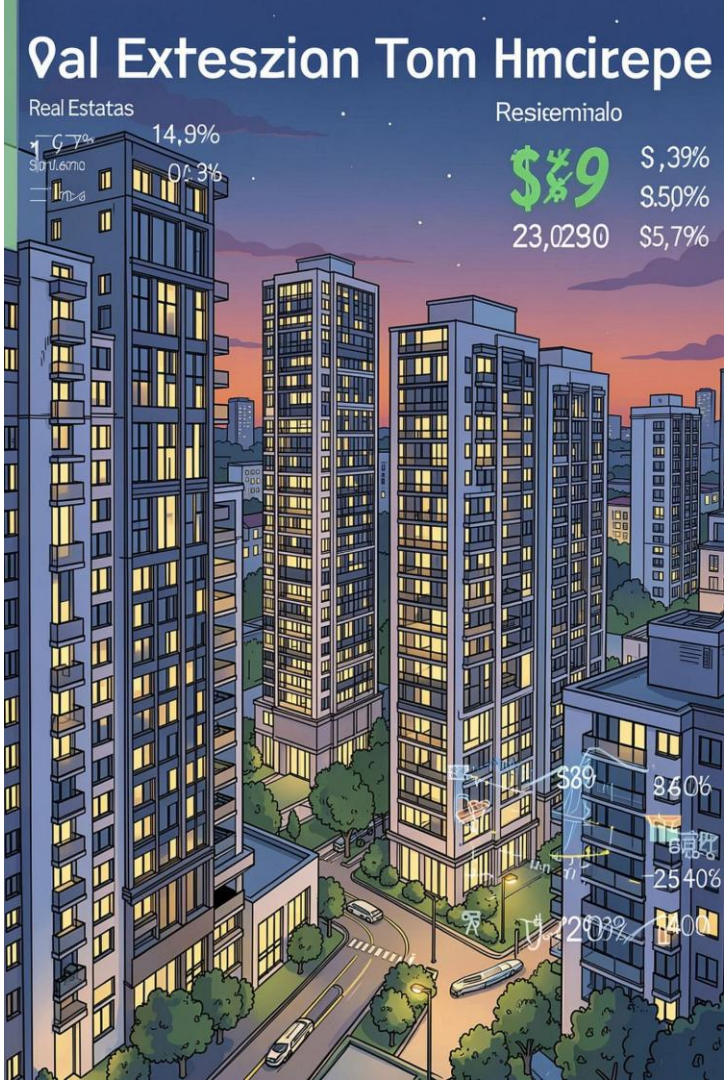
House Price Prediction System

Leveraging **Property Features, Location Intelligence, and Regression Models** to deliver accurate, data-driven residential property valuations for buyers, sellers, and real estate professionals.

MACHINE LEARNING | REAL ESTATE ANALYTICS | LOCATION INTELLIGENCE

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The Valuation Problem



Residential property valuation remains **highly manual, subjective, and inconsistent**. Multiple interdependent factors make accurate pricing a persistent challenge for all market participants.

Manual Assessments

Appraisals rely on expert opinion, introducing bias and variability.

Market Volatility

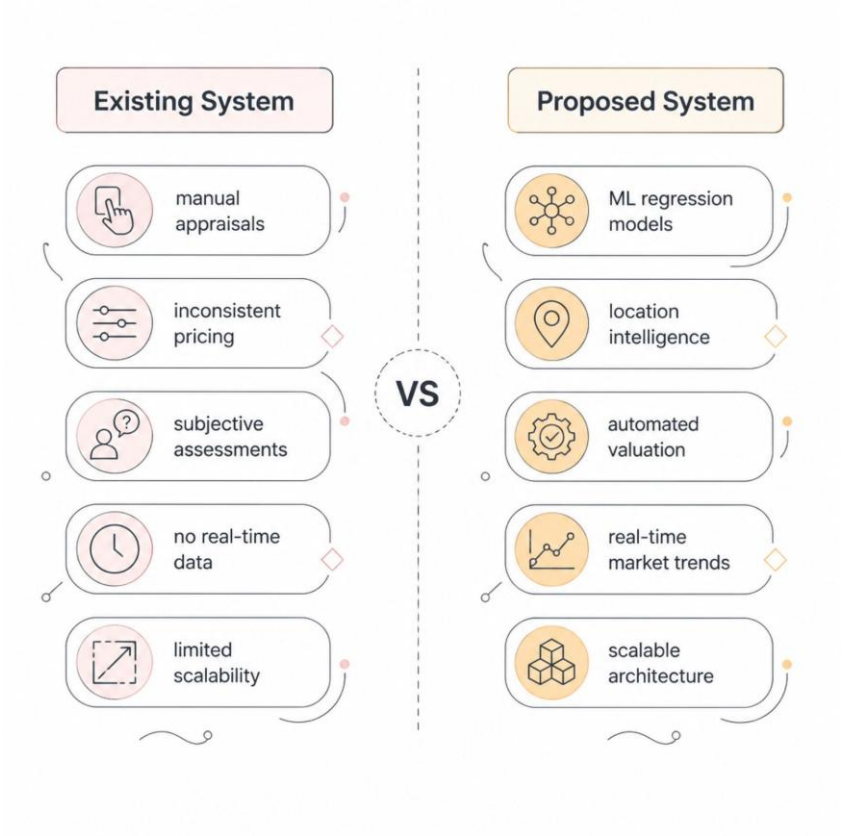
Rapid fluctuations render static valuations quickly outdated.

Multi-Factor Complexity

Location, size, amenities, and infrastructure interact in non-linear ways.

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Existing System vs. Proposed Solution



Why a New Approach?

Traditional methods cannot scale to meet modern real estate demands. Our proposed system replaces guesswork with **predictive intelligence** — analyzing thousands of historical transactions, property attributes, and geographic signals to estimate market value with statistical rigor.

The system is designed to assist — not replace — professional judgment, providing a reliable data-backed starting point for every valuation.

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System Objectives & Key Features



Accurate Valuation

Predict property prices using trained regression models with measurable error metrics.



Location Intelligence

Incorporate geographic factors — proximity to amenities, schools, and transit — into pricing.



Market Trend Analysis

Visualize historical price trends and comparative market analysis for informed decisions.



Multi-Role Access

Support Buyers, Sellers, Agents, and Administrators with tailored dashboards and tools.

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User Roles & System Access



Buyer

Search properties, view predicted prices, compare listings, and access market trend reports.



Seller

List properties, receive ML-based price estimates, and track listing performance.



Real Estate Agent

Manage multiple listings, generate client reports, and access comparative market analysis.

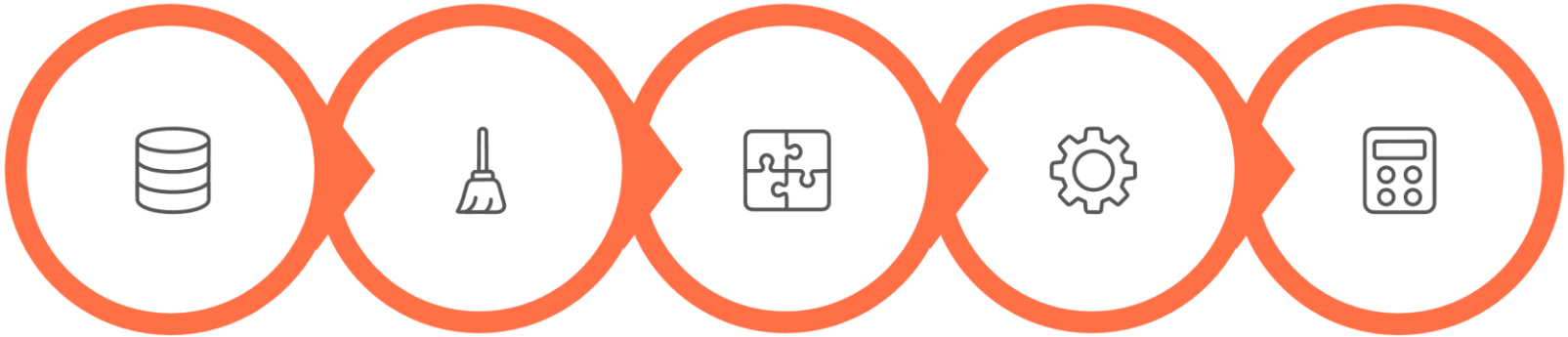


Administrator

Manage users, monitor system health, oversee ML model retraining, and configure settings.

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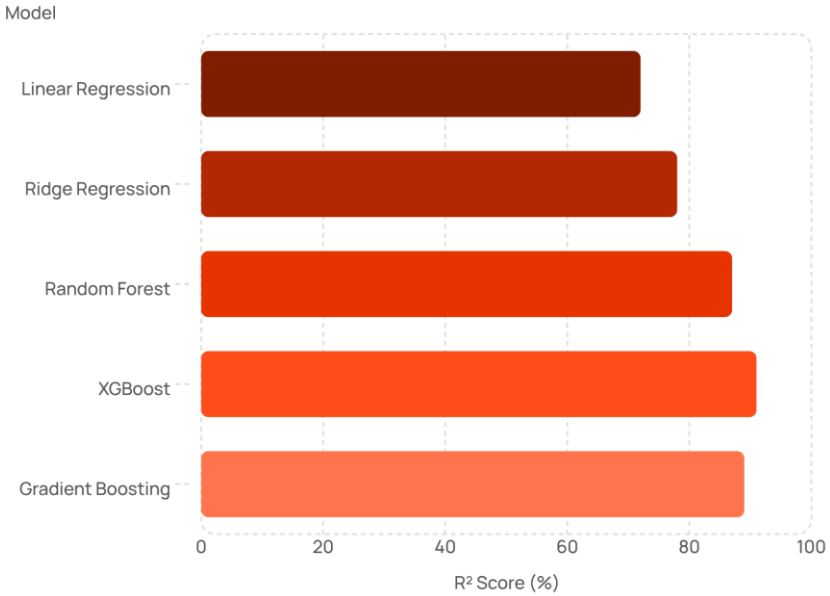
Data Pipeline & Machine Learning Workflow



The end-to-end pipeline transforms raw, heterogeneous real estate data into reliable price predictions — with each stage validated for data quality and model accuracy before production deployment.

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Regression Models & Performance Metrics



Evaluation Metrics

Model performance is measured across four key indicators to ensure prediction reliability:

XGBoost consistently outperforms other models, achieving the highest R^2 score on test data.

MAE

Mean Absolute Error — average prediction deviation

RMSE

Root Mean Squared Error — penalizes large errors

R^2 Score

Variance explained — model fit quality

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System Architecture & Data Flow



Three-Tier Architecture

Presentation Layer: React.js web interface with responsive dashboards for all user roles.

Application Layer: Python Flask/FastAPI backend hosting ML models, business logic, and REST APIs.

Data Layer: PostgreSQL for structured data, Redis for caching, and GeoServer for spatial queries.

→ DFD Level 0

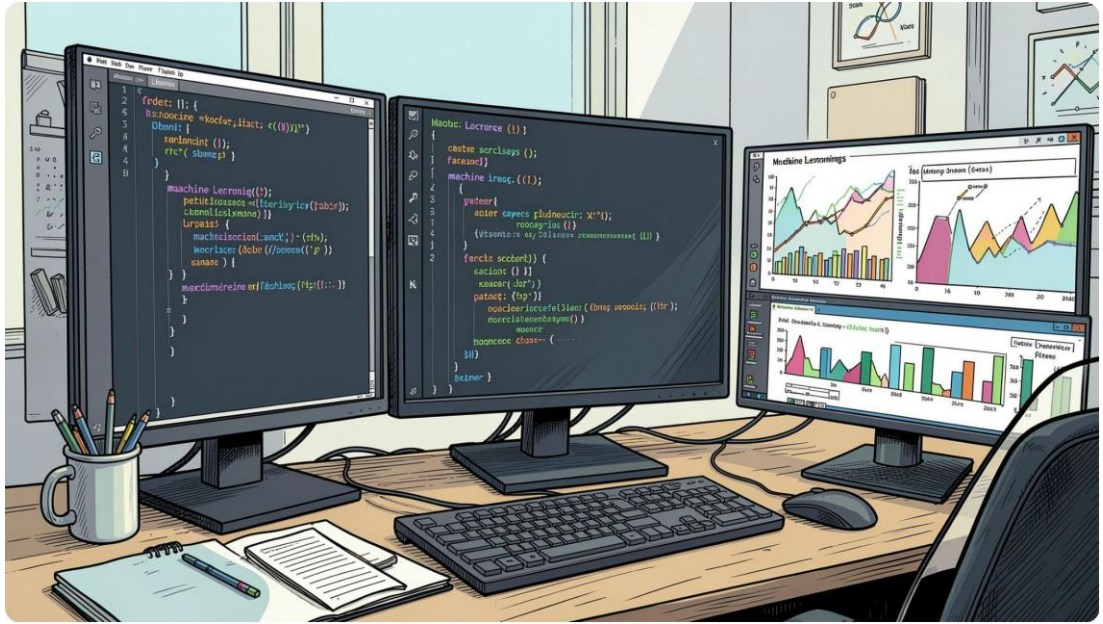
User → System → Prediction Output

→ DFD Level 1

Input validation → Feature extraction → Model inference → Report generation

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Technology Stack & Security



Core Technologies

Security Features

- JWT-based authentication with role-based access control
- End-to-end TLS encryption for data in transit
- Input validation and SQL injection prevention
- Audit logging for all administrative actions

Backend

Python, Flask/FastAPI, Scikit-learn, XGBoost, Pandas

Frontend

React.js, Chart.js, Leaflet.js (maps)

Database

PostgreSQL, PostGIS, Redis

Cloud & DevOps

AWS/GCP, Docker, CI/CD Pipelines

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Advantages, Applications & Future Roadmap

Key Advantages

- Reduced dependence on manual appraisals
- Improved pricing accuracy and transparency
- Scalable across cities and property types
- Real-time market trend visibility

Applications

- Residential property portals
- Bank mortgage and loan assessment
- Government land valuation systems
- Real estate investment analysis

Future Enhancements

- Deep learning with CNN for image-based features
- Real-time API integration with listing platforms
- Mobile application with AR property view
- Predictive analytics for investment forecasting

Expected Outcome: A production-ready, ML-powered valuation system that empowers stakeholders with accurate, transparent, and actionable property price insights — transforming how real estate decisions are made.

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